

§ 124. Reduced Numbers with Positive Discriminant

We now have to carry out a similar investigation for positive d , that is, for real quadratic irrational numbers. Such a number

$$\omega = x + y\sqrt{d}$$

shall be called reduced if it satisfies the following conditions:

ω is positive and greater than 1, and the number ω' conjugate to ω is negative and has absolute value less than 1.

In symbols,

$$0 < y\sqrt{d} - x < 1 < y\sqrt{d} + x. \quad (1)$$

If, in a plane, x, y are interpreted as rectangular coordinates, so that the points of the plane are the images of the numbers ω , then the images of the reduced numbers lie in the part of the plane shaded in Fig. 27, which corresponds to the assumption $d = 2$.

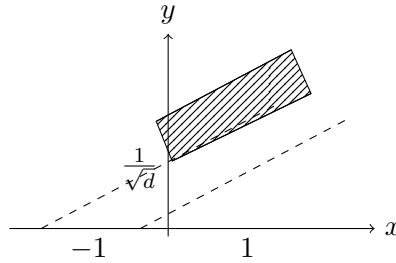


Fig. 27. The reduced region for $d = 2$.

We take proper and improper equivalence together here, and first prove the following theorems.

1. Every number ω is equivalent to a reduced number.

This is easily seen by expanding ω into a continued fraction and going as far as the complete quotient ω_n :

$$\omega = (\alpha_0, \alpha_1, \alpha_2, \dots, \alpha_{n-1}, \omega_n). \quad (2)$$

Then ω is equivalent to ω_n , and ω_n is positive and greater than 1 as soon as $n > 0$. Since ω_n is equivalent to ω , it remains only to show that, for sufficiently large n , the conjugate number ω'_n is negative and has absolute value less than 1. From (2), if $P_n : Q_n$ are the convergents, one has

$$\omega = \frac{P_n \omega_n + P_{n-1}}{Q_n \omega_n + Q_{n-1}},$$

and hence

$$\omega_n = -\frac{Q_{n-1}\omega - P_{n-1}}{Q_n\omega - P_n} = -\frac{Q_{n-1}}{Q_n} - \frac{(-1)^n}{Q_n(Q_n\omega - P_n)}. \quad (3)$$

Changing $\sqrt{d}$ into $-\sqrt{d}$ in this equation changes ω into ω' and ω_n into ω'_n ; consequently

$$\omega'_n = -\frac{Q_{n-1}\omega' - P_{n-1}}{Q_n\omega' - P_n} = -\frac{Q_{n-1}}{Q_n} \frac{\omega' - P_{n-1}/Q_{n-1}}{\omega' - P_n/Q_n}. \quad (4)$$

Furthermore

$$\omega'_n + 1 = \frac{1}{Q_n} \left\{ Q_n - Q_{n-1} - \frac{(-1)^n}{Q_n(\omega' - P_n/Q_n)} \right\}. \quad (5)$$

If n is taken sufficiently large, then

$$\omega' - \frac{P_n}{Q_n}, \quad \omega' - \frac{P_{n-1}}{Q_{n-1}}$$

differ arbitrarily little from the non-zero difference $\omega' - \omega = -2y\sqrt{d}$. Thus (4) shows that ω'_n is negative. From (5), however, it follows that $\omega'_n + 1$ is positive, since $Q_n - Q_{n-1}$ is a positive integer and therefore at least 1, while

$$\frac{1}{Q_n(\omega' - P_n/Q_n)}$$

becomes arbitrarily small for sufficiently large n . This proves theorem 1.

Since the conjugate of the conjugate ω' is the original number ω , the definition of reduced numbers also gives that ω and $-1 : \omega'$ are reduced simultaneously.

2. If ω is reduced and, for an integer α ,

$$\omega = \alpha + \frac{1}{\omega_1}, \quad (6)$$

then ω_1 is reduced if and only if α is the greatest integer contained in ω , that is, if

$$\alpha < \omega < \alpha + 1. \quad (7)$$

If (7) is fulfilled, then ω_1 is positive and greater than 1, and

$$\omega'_1 = \frac{1}{\omega' - \alpha} \quad (8)$$

is, since ω' is negative, a negative proper fraction. Thus ω_1 is reduced. Conversely, ω_1 cannot be reduced if $\alpha > \omega$, since then ω_1 would be negative, nor if $\alpha + 1 < \omega$, since then ω_1 would be less than 1.

Writing (8) in the form

$$-\frac{1}{\omega'_1} = \alpha + \frac{1}{-1/\omega'}, \quad (9)$$

we see that, since $-1 : \omega'$ and $-1 : \omega'_1$ are also reduced, α is the greatest integer contained in $-1 : \omega'_1$, and ω' is completely determined by ω'_1 from (9), hence ω is completely determined by ω_1 .

It follows that:

3. If a reduced number ω is developed into a continued fraction, then all its complete quotients ω_n are again reduced numbers; and by any one of these complete quotients ω_n both the following ω_{n+1} and the preceding ω_{n-1} are completely determined.

§ 125. Expansion of Real Quadratic Irrational Numbers into Continued Fractions

4. For a given discriminant there are only finitely many reduced numbers.

This is easily seen by writing the irrational number ω according to §. 122 in the form

$$\omega = \frac{b + \sqrt{D}}{2c} = \frac{-2a}{b - \sqrt{D}}, \quad D = b^2 + 4ac. \quad (1)$$

The conjugate number is

$$\omega' = \frac{b - \sqrt{D}}{2c} = \frac{-2a}{b + \sqrt{D}}. \quad (2)$$

For ω to be reduced one must have

$$0 < \frac{\sqrt{D} - b}{2c} < 1 < \frac{\sqrt{D} + b}{2c}. \quad (3)$$

Thus $\sqrt{D}$ must have the same sign as c ; taking it positive, this gives

$$0 < \sqrt{D} - b < 2c < \sqrt{D} + b. \quad (4)$$

A second form of the same conditions is obtained from the second representation:

$$0 < \sqrt{D} - b < 2a < \sqrt{D} + b. \quad (5)$$

Thus a, b, c are positive and $b < \sqrt{D}$.

Let λ be the greatest integer contained in $\sqrt{D}$, so that

$$\lambda < \sqrt{D} < \lambda + 1.$$

Then b has one of the values $1, 2, 3, \dots, \lambda$, with the restriction that if D is even only the even values, and if D is odd only the odd values, are to be used for b . The numbers a and c are then to be determined so that

$$\frac{D - b^2}{4} = ac, \quad (6)$$

and

$$\frac{\lambda - b + 1}{2} \leq c \leq \frac{\lambda + b}{2}, \quad (7)$$

while a must lie between the same bounds. Only one of the two endpoints is an integer; the other may be replaced, for the lower bound, by the next greater integer, and for the upper bound by the next smaller integer.

Thus only finitely many values of b are possible; for each of them $(D - b^2) : 4$ can be decomposed into two factors in only finitely many ways, and only those decompositions are retained for which both factors satisfy (7). One must also omit combinations in which a, b, c have a common divisor. This gives a concrete method for actually determining all reduced numbers for a given D .

We shall denote the numbers ω thus obtained by

$$\omega = \frac{\sqrt{D} + b}{2c} = \frac{2a}{\sqrt{D} - b} = \{a, b, c\}. \quad (8)$$

From 3. §. 124 and 4. §. 125 the following important theorem follows:

5. The continued-fraction expansion of a reduced number ω is periodic.

That is, if

$$\omega = (\alpha_0, \alpha_1, \alpha_2, \dots), \quad (9)$$

then, after a certain number of partial denominators, the same partial denominators recur in the same order. If the period consists of ν terms, then

$$\alpha_0 = \alpha_\nu = \alpha_{2\nu} = \dots, \quad \alpha_1 = \alpha_{\nu+1} = \alpha_{2\nu+1} = \dots, \quad \alpha_{\nu-1} = \alpha_{2\nu-1} = \alpha_{3\nu-1} = \dots.$$

Thus it suffices to denote the whole continued fraction by its period:

$$\omega = [\alpha_0, \alpha_1, \dots, \alpha_{\nu-1}]. \quad (10)$$

This theorem is clearly equivalent to the assertion that the complete quotient ω_ν of (9) is identical with ω itself, and this also contains the proof. All complete quotients ω_ν are equivalent numbers and therefore have the same discriminant; by 4. there are only finitely many possible complete quotients. Hence, in the sequence $\omega, \omega_1, \omega_2, \dots$, one of the complete quotients must occur a second time. If ω_ν is the first one which recurs, then it must be $\omega_\nu = \omega$, for otherwise, by 3. §. 124, the preceding complete quotient $\omega_{\nu-1}$ would already have occurred a second time.

The reduced numbers belonging to a discriminant can therefore be arranged in periods

$$\omega, \omega_1, \omega_2, \dots, \omega_{\nu-1}, \quad (11)$$

so that, if

$$\alpha_0, \alpha_1, \alpha_2, \dots, \alpha_{\nu-1}$$

are the greatest integers contained in them,

$$\omega = [\alpha_0, \alpha_1, \dots, \alpha_{\nu-1}], \quad \omega_1 = [\alpha_1, \alpha_2, \dots, \alpha_0],$$

and so on. If one period does not exhaust all reduced numbers, one forms a second period, and so forth. Since a number ω completely determines both its predecessor and its successor in the period, two different periods can have no number in common.

For these periods one has:

6. Reduced numbers from the same period are equivalent; reduced numbers from different periods are not equivalent.

The first part is immediate, since reduced numbers of the same period are complete quotients of one another. The second follows from §. 121, VI, according to which equivalent numbers, when their continued fractions are carried far enough, ultimately acquire the same complete quotients. Thus two periodic continued fractions of equivalent numbers must belong to the same period.

If, for a given discriminant D , the complete system of reduced numbers has been found by the preceding rules, it remains only to arrange them into periods and to find the partial denominators. Let

$$\omega = \frac{\sqrt{D} + b}{2c} = \frac{2a}{\sqrt{D} - b} \quad (12)$$

be one of these numbers and

$$\omega_1 = \frac{\sqrt{D} + b_1}{2c_1} = \frac{2a_1}{\sqrt{D} - b_1} \quad (13)$$

the number immediately following it in the period. Then

$$\omega = \alpha + \frac{1}{\omega_1}, \quad \omega_1 = \frac{1}{\omega - \alpha}, \quad (14)$$

where α is the greatest integer contained in ω . By (12) and (14),

$$\omega_1 = \frac{2c}{\sqrt{D} + b - 2c\alpha},$$

and comparison with (13) gives

$$a_1 = c, \quad b_1 = 2c\alpha - b. \quad (15)$$

Conversely, if two reduced numbers $\{a, b, c\}$ and $\{a_1, b_1, c_1\}$ belonging to D are related as in (15), for some integer α , then the second follows the first immediately in the period. Hence the number ω_1 is completely and uniquely determined from ω by (15).

7. Arrange the numbers $\{a, b, c\}$ from left to right so that the last number c of the preceding one is at the same time the first number a of the following one, and so that the sum of the two middle numbers $b + b_1$ is divisible by $2c$. This arrangement is possible in only one way, and the quotient $(b + b_1) : 2c$ is the number α that occurs as a partial denominator in the continued fraction.

Now consider any primitive integral quadratic equation

$$A + B\Omega + C\Omega^2 = 0, \quad (16)$$

where A, B, C are integers without a common divisor and satisfy

$$B^2 - 4AC = D.$$

Its two roots Ω, Ω' are quadratic irrational numbers with discriminant D . If we expand them into continued fractions, then by §. 124 we eventually reach complete quotients ω_1, ω_2 which are reduced, and which must therefore occur in the periods just described.

It remains to decide whether these two complete quotients occur in the same period or in different periods. By 6. they occur in the same period if and only if Ω and Ω' are equivalent. We shall call such quadratic irrational numbers, equivalent to their conjugates, two-sided numbers.¹ Thus the continued-fraction expansions of the roots of (16) lead to one or to two different periods according as the roots are two-sided or not. Which of the two cases occurs can be recognized from the period itself.

If ω is reduced and ω' is conjugate to ω , then $-1 : \omega'$ is also reduced. If Ω is equivalent to ω , then Ω' is equivalent to ω' , and hence also to $-1 : \omega'$ (§. 120). If

$$\omega = \{a, b, c\},$$

then, by (12),

$$-\frac{1}{\omega'} = \{c, b, a\}.$$

Thus the period for $-1 : \omega'$ is the reverse of the period for ω . In particular, if the period for ω is

$$[\alpha_0, \alpha_1, \dots, \alpha_{\nu-1}], \quad (17)$$

then that for $-1 : \omega'$ is

$$[\alpha_{\nu-1}, \alpha_{\nu-2}, \dots, \alpha_0]. \quad (18)$$

If ω is equivalent to ω' , hence also to $-1 : \omega'$, then by 6. the two periods must agree after beginning at a suitable place. Thus the period of ω must be reversible. In symbols, for some suitable k and for $i = 0, 1, 2, \dots, \nu - 1$,

$$\alpha_i = \alpha_{\nu+k-i}.$$

Conversely, if the period is reversible, then ω is equivalent to $-1 : \omega'$, and therefore to ω' . Thus:

8. Two-sided numbers have a reversible period in their continued-fraction expansion; conversely, numbers whose continued-fraction expansion has a reversible period are two-sided.

Finally, the property of having a periodic continued-fraction expansion distinguishes the real quadratic irrational numbers from all other real numbers. Indeed, if ω is expanded into a continued fraction and ω_n, ω_m are two complete quotients of this expansion, then

$$\omega = \frac{P_n\omega_n + P_{n-1}}{Q_n\omega_n + Q_{n-1}} = \frac{P_m\omega_m + P_{m-1}}{Q_m\omega_m + Q_{m-1}}.$$

If the continued fraction is periodic, then for two different values of n and m one has $\omega_n = \omega_m$; eliminating this quantity gives a quadratic equation for ω :

$$\begin{vmatrix} P_n - \omega Q_n & P_{n-1} - \omega Q_{n-1} \\ P_m - \omega Q_m & P_{m-1} - \omega Q_{m-1} \end{vmatrix} = 0.$$

¹Following Dedekind's suggestion, "two-sided" is used here instead of Gauss's *anceps* or the otherwise customary *ambig*; Dirichlet-Dedekind, *Vorlesungen über Zahlentheorie*, 4th ed., 1894, p. 139.

§ 126. Examples

We carry out several examples of the determination of reduced numbers and periods.

1. $D = 29$. Here $\lambda = 5$, and $b = 1, 3, 5$; hence

$$ac = \frac{D - b^2}{4} = 7, 5, 1.$$

The corresponding bounds for a and c are

$$3, 3; \quad 2, 4; \quad 1, 5.$$

The only reduced number is $\{1, 5, 1\}$, and

$$\omega = \frac{5 + \sqrt{29}}{2} = (5, 5, 5, \dots).$$

2. $D = 116 = 4 \cdot 29$. Here $\lambda = 10$,

$$b = 2, 4, 6, 8, 10, \quad ac = 28, 25, 20, 13, 4,$$

and the bounds are

$$5, 6; \quad 4, 7; \quad 3, 8; \quad 2, 9; \quad 1, 10.$$

After omitting the cases with common factor 2, the reduced numbers are

$$\{5, 4, 5\}, \quad \{4, 6, 5\}, \quad \{5, 6, 4\}, \quad \{1, 10, 4\}, \quad \{4, 10, 1\}.$$

They form the period

$$\{4, 10, 1\}, \{1, 10, 4\}, \{4, 6, 5\}, \{5, 4, 5\}, \{5, 6, 4\}, \{4, 10, 1\},$$

and

$$\frac{5 + \sqrt{29}}{2} = [10, 2, 1, 1, 2].$$

Starting with the fourth term gives $[1, 2, 10, 2, 1]$, the expansion of $(2 + \sqrt{29})/5$, and shows that the period is reversible.

3. $D = 76 = 4 \cdot 19$, $\lambda = 8$.

$$b = 2, 4, 6, 8, \quad ac = 18, 15, 10, 3.$$

The reduced numbers are

$$\{3, 4, 5\}, \{5, 4, 3\}, \{2, 6, 5\}, \{5, 6, 2\}, \{1, 8, 3\}, \{3, 8, 1\},$$

forming

$$\{3, 8, 1\}, \{1, 8, 3\}, \{3, 4, 5\}, \{5, 6, 2\}, \{2, 6, 5\}, \{5, 4, 3\}, \{3, 8, 1\}.$$

Thus

$$4 + \sqrt{19} = [8, 2, 1, 3, 1, 2],$$

again with a reversible period.

4. $D = 37$, $\lambda = 6$. The reduced numbers are

$$\{3, 1, 3\}, \quad \{1, 5, 3\}, \quad \{3, 5, 1\},$$

with period

$$\{3, 5, 1\}, \quad \{1, 5, 3\}, \quad \{3, 1, 3\}, \quad \{3, 5, 1\},$$

and

$$\frac{5 + \sqrt{37}}{2} = [5, 1, 1].$$

5. $D = 148 = 4 \cdot 37$, $\lambda = 12$.

$$b = 2, 4, 6, 8, 10, 12, \quad ac = 36, 33, 28, 21, 12, 1.$$

The reduced numbers are

$$\{4, 6, 7\}, \{7, 6, 4\}, \{3, 8, 7\}, \{7, 8, 3\}, \{3, 10, 4\}, \{4, 10, 3\}, \{1, 12, 1\}.$$

They split into three periods:

$$\begin{aligned} &\{1, 12, 1\}, \{1, 12, 1\}, \\ &\{3, 10, 4\}, \{4, 6, 7\}, \{7, 8, 3\}, \{3, 10, 4\}, \\ &\{4, 10, 3\}, \{3, 8, 7\}, \{7, 6, 4\}, \{4, 10, 3\}. \end{aligned}$$

The continued-fraction periods are

$$[12], \quad [2, 1, 3], \quad [3, 1, 2].$$

6. $D = 136 = 4 \cdot 34$, $\lambda = 11$.

$$b = 2, 4, 6, 8, 10, \quad ac = 33, 30, 25, 18, 9.$$

The periods obtained from the reduced numbers are

$$\{5, 4, 6\}, \{6, 8, 3\}, \{3, 10, 3\}, \{3, 8, 6\}, \{6, 4, 5\}, \{5, 6, 5\}, \{5, 4, 6\},$$

and

$$\{9, 10, 1\}, \{1, 10, 9\}, \{9, 8, 2\}, \{2, 8, 9\}, \{9, 10, 1\}.$$

The corresponding continued-fraction periods are

$$[1, 3, 3, 1, 1, 1], \quad [10, 1, 4, 1].$$

Thus here there are two different periods, both reversible.

§ 127. Pell's Equation

Let ω be a reduced number of discriminant D . From its continued-fraction expansion,

$$\omega = \frac{P_n \omega_n + P_{n-1}}{Q_n \omega_n + Q_{n-1}}, \tag{1}$$

and $\omega_n = \omega$ whenever n is a multiple of the length ν of the period. It follows that

$$Q_n \omega^2 = (P_n - Q_{n-1})\omega + P_{n-1}. \tag{2}$$

Let ω satisfy

$$c\omega^2 = a + b\omega, \quad D = b^2 + 4ac. \quad (3)$$

Since a, b, c have no common divisor, (2) must be obtained from (3) by multiplication by an integer u :

$$Q_n = uc, \quad P_{n-1} = ua, \quad P_n - Q_{n-1} = ub. \quad (4)$$

Put

$$P_n + Q_{n-1} = t. \quad (5)$$

Then

$$P_n = \frac{t + ub}{2}, \quad Q_n = uc, \quad P_{n-1} = ua, \quad Q_{n-1} = \frac{t - ub}{2}. \quad (6)$$

Using

$$P_n Q_{n-1} - Q_n P_{n-1} = (-1)^n$$

one obtains

$$t^2 - Du^2 = (-1)^n 4. \quad (7)$$

This is Pell's equation. The problem is to find all integral solutions t, u for a given D . Since changing signs gives new solutions, it suffices to determine the positive solutions.

The formulae (4), (5) give infinitely many such solutions: choose n to be any multiple of the period length ν , take u as the greatest common divisor of $Q_n, P_{n-1}, P_n - Q_{n-1}$, or simply as $Q_n : c$, and then determine t from (5) or from (7).

If ν is odd and n is an odd multiple of ν , then

$$t^2 - Du^2 = -4; \quad (8)$$

if ν is even, or n an even multiple of ν , then

$$t^2 - Du^2 = 4. \quad (9)$$

Equation (9) has the trivial solution $t = 2, u = 0$. Apart from that case, the solutions considered here are positive.

Conversely, every positive solution of (8) or (9) is obtained in this way. Let t, u be such a solution and let $\omega = \{a, b, c\}$ be a reduced irrational number of discriminant D . Define $P_n, Q_n, P_{n-1}, Q_{n-1}$ by (6). These are integers, because t and ub are either both even or both odd. Equation (3) then implies (2), hence

$$\omega = \frac{P_n \omega + P_{n-1}}{Q_n \omega + Q_{n-1}}. \quad (10)$$

From (8) or (9) one has

$$P_n Q_{n-1} - Q_n P_{n-1} = \mp 1, \quad (11)$$

where the upper or lower sign is used according as t, u satisfy (8) or (9). From (6),

$$Q_n - Q_{n-1} = \frac{(2c + b)u - t}{2},$$

and since $\{a, b, c\}$ is reduced, $2c + b > \sqrt{D}$, whence

$$Q_n - Q_{n-1} > \frac{u\sqrt{D} - t}{2} = \frac{u^2 D - t^2}{2(u\sqrt{D} + t)} = \frac{\pm 2}{u\sqrt{D} + t}.$$

Thus $Q_{n-1} \leq Q_n$, equality being possible only in the case of (9). Similarly, because $b < \sqrt{D}$,

$$Q_{n-1} = \frac{t - ub}{2} > \frac{t - u\sqrt{D}}{2} = \frac{t^2 - u^2 D}{2(t + u\sqrt{D})} = \frac{\mp 2}{t + u\sqrt{D}},$$

so

$$0 \leq Q_{n-1} \leq Q_n. \quad (12)$$

Develop $P_n : Q_n$ into a finite continued fraction

$$\frac{P_n}{Q_n} = (\alpha_0, \alpha_1, \dots, \alpha_{n-1})$$

with n chosen so that $(-1)^n = \mp 1$. If $P' : Q'$ is the preceding convergent, then

$$P_n Q' - Q_n P' = \mp 1, \quad 0 \leq Q' \leq Q_n.$$

By §. 118, III, this gives $Q' = Q_{n-1}$ and $P' = P_{n-1}$. Hence

$$(\alpha_0, \alpha_1, \dots, \alpha_{n-1}, \omega) = \omega,$$

so $[\alpha_0, \alpha_1, \dots, \alpha_{n-1}]$ is a period for ω , and the solution t, u is precisely the one obtained from (4), (5). This proves that all solutions of (7) are obtained by the continued-fraction method.

The least positive solution will be denoted by T, U . It is obtained by taking n as small as possible, namely $n = \nu$. Hence if ν is even only (9) is soluble, whereas if ν is odd both (8) and (9) are soluble. Since this depends only on the discriminant D , it follows that:

9. For one discriminant the different periods all contain either an even number of terms or all contain an odd number of terms.

If D is even, hence divisible by 4, then t is even and (7) can be divided term by term by 4. If D is odd, then t, u are either both even or both odd; in the latter case their squares are both congruent to 1 modulo 8, so $D \equiv 5 \pmod{8}$ is necessary. It is not, however, always possible when $D \equiv 5 \pmod{8}$ to solve (7) by odd numbers.² If $D \equiv 1 \pmod{8}$, then t, u must be even.

For the examples of §. 126 the smallest positive solutions are:

$$\begin{aligned} 5^2 - 29 \cdot 1^2 &= -4, \\ 7^2 - 29 \cdot 13^2 &= -1, \\ 17^2 - 19 \cdot 39^2 &= 1, \\ 6^2 - 37 \cdot 1^2 &= -1, \\ 35^2 - 34 \cdot 6^2 &= 1. \end{aligned}$$

§ 128. Derivation of All Solutions of Pell's Equation from the Least Positive One

For any solution t, u of Pell's equation, the expression

$$\frac{t^2 - Du^2}{4}$$

is the norm of the conjugate numbers

$$\frac{t + u\sqrt{D}}{2}, \quad \frac{t - u\sqrt{D}}{2}.$$

We call them the units belonging to $\sqrt{D}$. If (t_1, u_1) and (t_2, u_2) are two solutions, then

$$\Theta_1 = \frac{t_1 + u_1\sqrt{D}}{2}, \quad \Theta_2 = \frac{t_2 + u_2\sqrt{D}}{2}$$

²Cf. Cayley, Note sur l'équation $x^2 - Dy^2 = \pm 4$, $D \equiv 5 \pmod{8}$, Crelle's Journal, vol. 53; *Mathematical Papers*, vol. IV. Tables for solutions of Pell's equation in the form $y^2 = ax^2 + 1$ were computed by Degen (*Canon Pellianus*, Copenhagen 1817). Such a table is also found in Legendre's number theory, German translation by Maser.

have norm ± 1 , and so does their product

$$\Theta_3 = \Theta_1 \Theta_2 = \frac{t_3 + u_3 \sqrt{D}}{2},$$

where

$$t_3 = \frac{t_1 t_2 + u_1 u_2 D}{2}, \quad u_3 = \frac{t_1 u_2 + t_2 u_1}{2}.$$

Thus the product of two units is again a unit.

If Θ is any unit, then $\pm \Theta^{-1}$ is its conjugate unit. Except for ± 1 , the units therefore occur four at a time,

$$\pm \Theta, \quad \pm \Theta^{-1},$$

one of which is positive and greater than 1. If T, U is the least positive solution, then

$$\Theta = \frac{T + U \sqrt{D}}{2}$$

is the least positive unit belonging to $\sqrt{D}$ that is greater than 1. All units belonging to $\sqrt{D}$ are contained in the form $\pm \Theta^n$, where n is an integer. It is enough to show this for units greater than 1. If Θ_1 is such a unit, then it lies between two consecutive powers of Θ :

$$\Theta^n \leq \Theta_1 < \Theta^{n+1}.$$

Hence

$$1 \leq \Theta_1 \Theta^{-n} < \Theta.$$

But $\Theta_1 \Theta^{-n}$ is again a unit; unless equality holds, it is a unit greater than 1 and less than Θ , contrary to the minimality of Θ . Therefore $\Theta_1 = \Theta^n$.

Remark concerning Gauss's theory of quadratic forms.

To connect the preceding considerations with the usual Gauss theory of quadratic forms, Weber adds the following observations. Two quadratic forms

$$(A, B, C), \quad (-A, B, -C)$$

are equivalent in Gauss's sense if and only if the period of the continued fraction has an odd number of terms. A reduced number $\omega = \{a, b, c\}$ is the first root of

$$(a, \tfrac{1}{2}b, -c) \quad \text{or} \quad (2a, b, -2c),$$

according as D is even or odd, and the second root of

$$(-a, \tfrac{1}{2}b, c) \quad \text{or} \quad (-2a, b, 2c).$$

If these forms are equivalent, so that the periods of ω have odd length, then each two-sided class of forms corresponds to one system of equivalent numbers ω , and each pair of opposite classes corresponds to two such systems. Hence the class number in Gauss's sense is equal to the number of periods of reduced numbers. If the period length is even, the two forms are not equivalent, and the Gauss class number is twice the number of periods.

§ 129. Approximate Calculation of the Real Roots of a Numerical Equation by Continued Fractions

The theory of continued fractions gives a method due to Lagrange for calculating the real roots of a numerical equation to arbitrary accuracy. Let $f(x) = 0$ be a real equation with at least one real root. One first finds an integer α_0 such that between α_0 and $\alpha_0 + 1$ lies one or more roots of $f(x)$. Transform $f(x)$ by

$$x = \alpha_0 + \frac{1}{x_1}$$

into $f_1(x_1)$; then f_1 has one or more roots greater than 1. Choose α_1 so that between α_1 and $\alpha_1 + 1$ lies one or more roots of $f_1(x_1)$, and continue in the same way. Eventually one isolates a single root and obtains the convergents

$$(\alpha_0), \quad (\alpha_0, \alpha_1), \quad (\alpha_0, \alpha_1, \alpha_2), \quad \dots$$

which alternately lie above and below the root and converge to it without bound. If

$$(\alpha_0, \alpha_1, \dots, \alpha_{\nu-1}) = \frac{P_{\nu}}{Q_{\nu}},$$

then the error is less than $1/Q_{\nu}^2$.

For example, take

$$x^3 - 2x - 2 = 0.$$

It has a root between 1 and 2, so $\alpha_0 = 1$. The transformations give

$$\begin{aligned} x^3 - 2x - 2 &= 0, & \alpha_0 &= 1, \\ 3x^3 - x^2 - 11x - 1 &= 0, & \alpha_1 &= 1, \\ 2x^3 - 4x^2 - 8x - 3 &= 0, & \alpha_2 &= 3, \\ 9x^3 - 22x^2 - 11x - 2 &= 0, & \alpha_3 &= 2, \\ 43x^3 - 6x^2 - 32x - 9 &= 0, & \alpha_4 &= 1, \\ x^3 - 94x^2 - 132x - 46 &= 0, & \alpha_5 &= 95, \\ 3561x^3 - 9183x^2 - 191x - 1 &= 0. \end{aligned}$$

Thus

$$x = (1, 1, 3, 2, 1, 95, \dots),$$

with convergents

$$1, \quad \frac{2}{1}, \quad \frac{7}{4}, \quad \frac{16}{9}, \quad \frac{23}{13}, \quad \frac{2201}{1244}, \quad \frac{4425}{2501}.$$

The last two give the bounds

$$1.7692926, \quad 1.76929228,$$

and the error of the latter, slightly too small, is less than 0.00000016. The rapid convergence here is due to the relatively large partial denominator 95, which can be foreseen roughly from the negative coefficient of the second-highest power in the preceding transformed equation.

§ 130. Rational Roots of Integral Equations. Reducible Equations

We end the section with a consideration of integral equations. For an equation with rational coefficients one first clears denominators. One may also reduce to the case of a monic equation

$$x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n = 0 \tag{1}$$

with integral coefficients. A rational root of (1) cannot be a proper fraction. If $x = p/q$ with $(p, q) = 1$ and $q > 1$, multiplication by q^{n-1} would make p^n/q an integer, which is impossible. Therefore any rational root is an integer divisor of a_n .

The same question leads to the more general problem of deciding whether an integral polynomial

$$f(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n$$

is divisible by a lower-degree rational polynomial

$$\varphi(x) = x^\nu + \alpha_1x^{\nu-1} + \alpha_2x^{\nu-2} + \cdots + \alpha_{\nu-1}x + \alpha_\nu.$$

If f is divisible by φ , the quotient is again a rational polynomial, and it suffices to consider $\nu \leq n/2$. The coefficients of both factors must in fact be integers. Dividing $f(x)$ by $\varphi(x)$ with undetermined coefficients gives a remainder of degree $\nu - 1$; setting its coefficients equal to zero gives ν equations for the unknown α 's. These may be eliminated, but often it is better to solve the system directly in integers.

As an example, Weber considers

$$x^{10} - 36x^8 + 528x^6 - 3897x^4 + 14354x^2 - 21025 = 0,$$

from the theory of elliptic functions.³ Suppose one factor is

$$x^5 + \alpha x^4 + \beta x^3 + \gamma x^2 + \delta x + \varepsilon.$$

The other must then be

$$x^5 - \alpha x^4 + \beta x^3 - \gamma x^2 + \delta x - \varepsilon.$$

Equating coefficients gives

$$\begin{aligned} \alpha^2 - 2\beta &= 36, \\ \beta^2 + 2\delta - 2\alpha\gamma &= 528, \\ \gamma^2 + 2\alpha\varepsilon - 2\beta\delta &= 3897, \\ \delta^2 - 2\gamma\varepsilon &= 14354, \\ \varepsilon^2 &= 21025. \end{aligned}$$

Thus $\varepsilon = 145$. From the fourth equation modulo 145 one obtains

$$\delta^2 \equiv -1 \pmod{145},$$

with solutions $\delta = \pm 12, \pm 17 \pmod{145}$; since δ must be even, the relevant preliminary values are

$$\delta = \pm 12, \pm 128, \pm 162.$$

These give $\gamma = -49, 7, 41$. A further reduction modulo 5 excludes all but

$$\delta = -12, +128, -162,$$

and then the middle case, and also $\delta = -12$, are excluded. Hence

$$\delta = -162, \quad \gamma = 41.$$

The third equation gives

$$17\beta \equiv 93 \pmod{145},$$

hence $\beta \equiv 14 \pmod{145}$. Taking $\beta = 14$ gives $\alpha = -8$, and the factorisation is

$$\begin{aligned} &x^{10} - 36x^8 + 528x^6 - 3897x^4 + 14354x^2 - 21025 \\ &= (x^5 - 8x^4 + 14x^3 + 41x^2 - 162x + 145)(x^5 + 8x^4 + 14x^3 - 41x^2 - 162x - 145). \end{aligned}$$

³Cf. Weber, *Elliptische Functionen und algebraische Zahlen*, Braunschweig 1891; the example is from the transformation theory of degree 47, *Mathematische Annalen*, vol. 43.